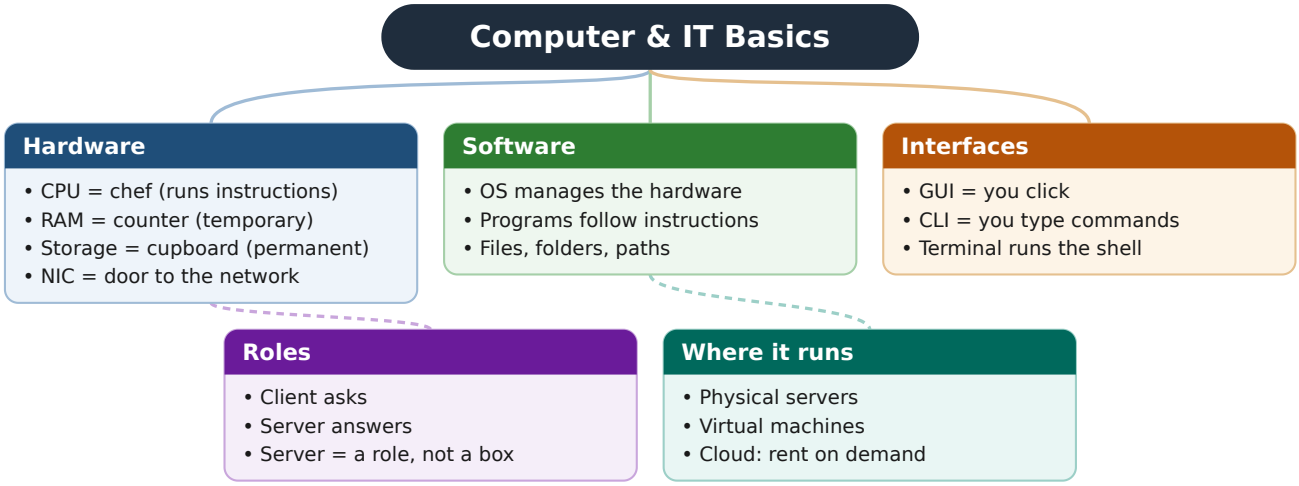
1. Computer & IT Basics

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In this section: Core parts of a computer · Key terms · First Windows PowerShell commands



Core parts of a computer

Part	What it does	Remember it as
CPU	Runs instructions and calculations	The chef
RAM	Fast, short-term working memory. Cleared at power off	The kitchen counter
Storage (SSD/HDD)	Keeps files permanently	The cupboard
Network card (NIC)	Connects the computer to a network	The front door

Key terms

Term	Meaning
Operating system (OS)	The main program that manages hardware and runs apps (Windows, macOS, Linux)
Program / application	A set of instructions the computer follows
File / folder (directory)	Saved data / a container for files
Path	The address of a file. Windows: `C:\Users\me\file.txt`. Linux: `/home/me/file.txt`
GUI	Graphical interface: you click
CLI / terminal / shell	Text interface: you type commands
Client	The computer or program that asks for something
Server	The computer or program that answers. "Server" is a role, not a special machine
Virtual machine (VM)	A full computer simulated in software, running on a real one
Cloud	Renting computers, storage and services over the internet (AWS, Azure, GCP)

Why DevOps uses the terminal: servers have no screen, commands can be automated and repeated, and they are exact and fast.

First Windows PowerShell commands

Command	What it does
`hostname`	Shows the computer's name
`whoami`	Shows the logged-in user
`ipconfig`	Shows your IP address and gateway
`ipconfig /all`	Full network details, including the MAC address and DNS servers
`ping google.com`	Tests whether another computer can be reached
`cls`	Clears the screen

REMEMBER

- Every computer: **Input → Process → Store → Output**.
- RAM is fast but forgets at power-off; storage keeps data.
- Servers usually run **Linux**, have no screen, and are managed through a **terminal** over a network.